

Circuit Analysis (Lab11)

Experiment 10: Verification of Norton's Theorems

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Date: 13/ May/26 Grade and Signature:

CLO2 CLO3	Use with identification the circuit components, breadboards, multimeter, power supplies, signal generators, and oscilloscopes. Make measurements in their own constructed electric circuits.			
Psychomotor/Affective	Level1 (1)	Level2 (2-3)	Level3 (4-5)	Level4 (6-7)
Report Marks (3)				
	Total marks (10)			

Objective

The objective of this lab is verify Nortan's theorem and to find the full load current.

Apparatus Required

Sr. no	Apparatus	Range	Quantity
1	Regulated Power Supply	Standard	1
2	Signal Generator	Standard	1
3	Oscilloscope	Standard	1
4	Digital Multimeter	Standard	1
5	Breadboard & Wires	Standard	As Required
6	Resistors	1k, 330 ohms	1,3
7	Potentiometer	Standard	1

Description

Statement:

Any linear circuit can be simplified to an equivalent circuit consisting of a single current source and parallel resistance that is connected to a load.

Precautions:

1. Voltage control knob of RPS should be kept at minimum position.
2. Current control knob of RPS should be kept at maximum position

Procedure for Norton's Theorem:

1. Connections are given as per circuit diagram.
2. Set a particular value in RPS and note down the ammeter readings in the original circuit.

To Find I_N :

3. Remove the load resistance and short circuit the terminals.
4. For the same RPS voltage note down the ammeter readings.

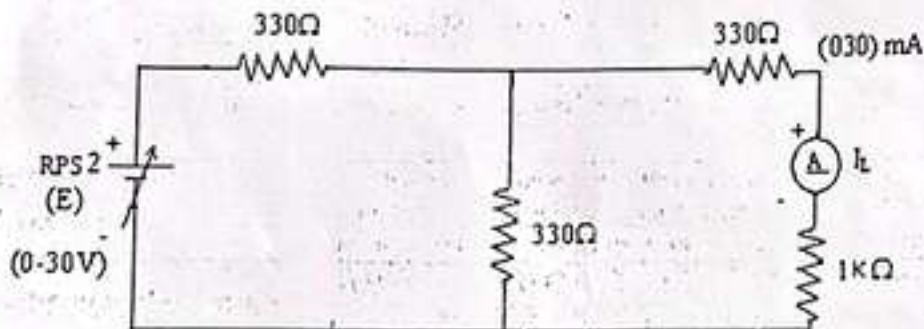
To Find R_N :

5. Remove RPS and short circuit the terminal and remove the load and note down the resistance across the two terminals.

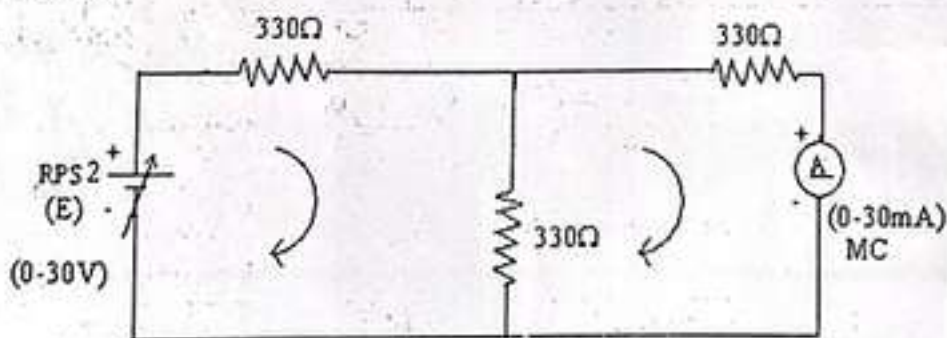
Equivalent Circuit:

6. Set I_N and R_N and note down the ammeter readings.
7. Verify Norton's theorem.

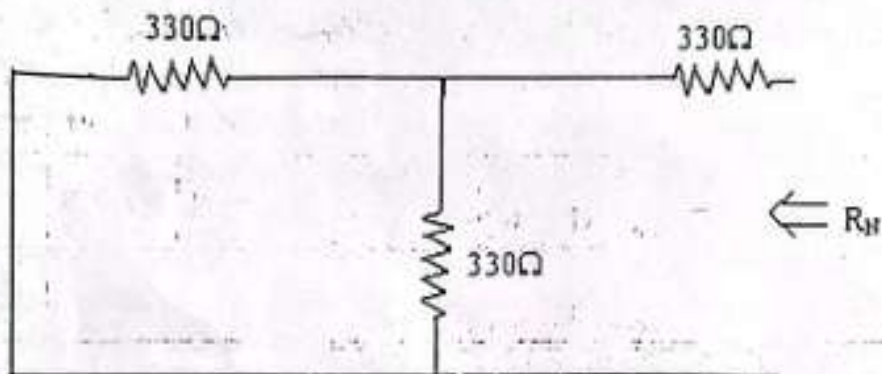
To find load current in circuit 1:



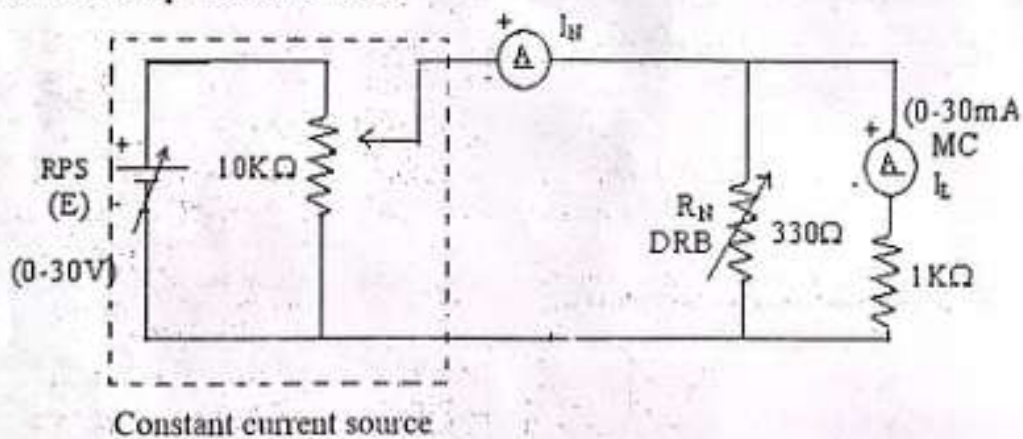
To find I_N



To find R_N



Norton's equivalent circuit



Norton's Practical Values

	E (volts)	I_N (mA)	R_N (ohms)	I_L (mA)	
				Circuit - I	Equivalent Circuit
Practical Values	30V	30.3mA	495.00	10.0mA	10.0mA

Post Lab Questions

1. What is Norton's Theorem?

Any linear electrical circuit can be replaced by an equivalent current source in parallel with a resistance. It simplifies circuit analysis. The current source is called Norton current (I_N). The parallel resistance is called Norton Resistance (R_N). It is widely used in network analysis and circuit design.

2. What are the applications of Norton's Theorem?

- 1) It simplifies complex electrical circuit into a simpler equivalent circuit.
- 2) It helps in finding current through different load resistors.
- 3) It is used in circuit analysis and design.
- 4) It makes troubleshooting and testing of electrical networks easier.
- 5) It is useful in electronic and power system applications.

3. What are the theoretical and practical limitations of Norton's Theorem?

- 1) It is applicable only to linear electrical circuits.
- 2) It does not work directly with non-linear components like diodes and transistors.
- 3) It becomes difficult when circuits contain many dependent source.
- 4) In practical circuits, internal losses and component tolerances can affect accuracy.
- 5) Finding Norton current (I_N) and resistance in complex circuits may take extra time.